

```
!pip install diffusers transformers torch accelerate
```

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Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from diffusers) (3.29.0)
Requirement already satisfied: httpx<1.0.0 in /usr/local/lib/python3.12/dist-packages (from diffusers) (0.28.1)
Requirement already satisfied: huggingface-hub<2.0,>=0.34.0 in /usr/local/lib/python3.12/dist-packages (from diffusers) (1.11.4)
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Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from diffusers) (2025.11.3)
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Requirement already satisfied: Pillow in /usr/local/lib/python3.12/dist-packages (from diffusers) (11.3.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.1)
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Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
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Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.93)
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Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=0.34.0->diffusers) (0.0.2)
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```

```
from diffusers import StableDiffusionPipeline
import torch
from IPython.display import display

print("Loading the model into the GPU... this will take about 1-2 minutes the first time.")

# Load the free Stable Diffusion model. We use float16 to save memory on the free GPU.
pipe = StableDiffusionPipeline.from_pretrained(
    "runwayml/stable-diffusion-v1-5",
    torch_dtype=torch.float16
)

# Move the model to the graphics card
pipe = pipe.to("cuda")

print("\n🤖 Free Text-to-Image Bot Ready! Type 'exit' to stop.\n")

while True:
    user_input = input("You: ")
```



```
if user_input.lower() == "exit":
    print("Shutting down the bot. Great job!")
    break

print("Bot: Generating your image... please wait a few seconds!")

# Generate the image based on the prompt
image = pipe(user_input).images[0]

# Display the image directly in the Colab output
display(image)
print("\n")
```



Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recommend migrating to PyTorch classes or pinning your v
Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recommend migrating to PyTorch classes or pinning your v
Loading the model into the GPU... this will take about 1-2 minutes the first time.
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
model_index.json: 100% 541/541 [00:00<00:00, 57.9kB/s]
Download complete: 5.48G/? [00:53<00:00, 130MB/s]
Fetching 15 files: 100% 15/15 [00:43<00:00, 3.21s/it]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
Loading pipeline components...: 100% 7/7 [00:25<00:00, 3.26s/it]
Loading weights: 100% 396/396 [00:04<00:00, 60.93it/s, Materializing param=visual_projection.weight]
StableDiffusionSafetyChecker LOAD REPORT from: /root/.cache/huggingface/hub/models--runwayml--stable-diffusion-v1-5/snapshots/45
Key | Status | |
-----+-----+--
vision_model.vision_model.embeddings.position_ids | UNEXPECTED | |
Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Loading weights: 100% 196/196 [00:02<00:00, 119.93it/s, Materializing param=text_model.final_layer_norm.weight]
CLIPTextModel LOAD REPORT from: /root/.cache/huggingface/hub/models--runwayml--stable-diffusion-v1-5/snapshots/451f4fe16113bff5a
Key | Status | |
-----+-----+--
text_model.embeddings.position_ids | UNEXPECTED | |
Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
🤖 Free Text-to-Image Bot Ready! Type 'exit' to stop.
You: panda eating bambao
Bot: Generating your image... please wait a few seconds!
100% 50/50 [00:07<00:00, 7.60it/s]

